

Xinyao Li

PHD CANDIDATE

School of Computer Science & Engineering, University of Electronic Science and Technology of China (UESTC)

✉ xinyao326@std.uestc.edu.cn | 🏠 xinyao326.github.io | 📄 Google Scholar

Education

University of Electronic Science and Technology of China

BACHELOR DEGREE IN COMPUTER SCIENCE

2018 - 2022

- Advisor: Prof. Jingjing Li

University of Electronic Science and Technology of China

MASTER CANDIDATE IN COMPUTER SCIENCE

2022 - 2023

- Advisor: Prof. Jingjing Li

University of Electronic Science and Technology of China

PHD CANDIDATE IN MACHINE LEARNING/ARTIFICIAL INTELLIGENCE

2023 - present

- Advisor: Prof. Jingjing Li
- In Successive Postgraduate and Doctoral Program

Agency for Science, Technology and Research (A*STAR)

VISITING PHD STUDENT

Singapore

2025.8 - 2026.8

- Co-advised by: Prof. Ivor Tsang and Dr. Hangwei Qian

Publications

X. Li, H. Qian, J. Li, L. Zhu, and I. Tsang, “**Learning Ordinality-Aware Multimodal Representations for Composite Materials Design**,” *Nature Communications*, 2026

X. Li, J. Li, Z. Du, L. Zhu, and H. T. Shen, “**Unified Modality Separation: A Vision-Language Framework for Unsupervised Domain Adaptation**,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025

X. Li, Y. Li, Z. Du, F. Li, K. Lu, and J. Li, “**Split to Merge: Unifying Separated Modalities for Unsupervised Domain Adaptation**,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024

X. Li, Y. Min, H. Chen, Z. Du, F. Li, and J. Li, “**Generalizing Vision-Language Models with Dedicated Prompt Guidance**,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026

X. Li, J. Li, F. Li, L. Zhu, and K. Lu, “**Agile Multi-Source-Free Domain Adaptation**,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024

X. Li, D. Zhang, Z. Du, L. Zhu, Z. Chen, and J. Li, “**PatAug: Augmentation of Augmentation for Test-Time Adaptation**,” in *Proceedings of the 33rd ACM international conference on multimedia*, 2025

X. Li, Z. Du, J. Li, L. Zhu, and K. Lu, “**Source-Free Active Domain Adaptation via Energy-Based Locality Preserving Transfer**,” in *Proceedings of the 30th ACM international conference on multimedia*, 2022

X. Li, J. Li, F. Li, L. Zhu, Y. Yang, H. T. Shen, “**Generalizing Vision-Language Models to Novel Domains: A Comprehensive Survey**,” *ACM Transactions on Multimedia Computing, Communications and Applications*, 2026

X. Li, J. Li, L. Zuo, L. Zhu, and H. T. Shen, “**Domain Adaptive Remaining Useful Life Prediction with Transformer**,” *IEEE Transactions on Instrumentation and Measurement*, 2022

X. Li, J. Li, L. Zhu, and H. T. Shen, “**Efficient Transfer Learning of Large Models with Limited Resources: A Survey**,” *Chinese Journal of Computers*, 2024

X. Li, H. Chen, L. Shen, X. Feng, and J. Li, “**Cross-Domain State Estimation of Lithium-ion Batteries: A Review**,” *Journal of University of Electronic Science and Technology of China*, 2024

S. Lv*, X. Li*, J. Li, “**CONGA: Confidence-and-Gradient-Aware Learning Rate Schedule for Test Time Adaptation**,” in *International Conference on Machine Learning*, 2026

Z. Du, X. Li, F. Li, K. Lu, L. Zhu, and J. Li, “**Domain-Agnostic Mutual Prompting for Unsupervised Domain Adaptation**,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024

Honors & Awards

2024, 2025 **Doctoral National Scholarship,**
UESTC Academic Scholarship, First Prize

top 3%
top 10%

Research Interests

Transfer learning Domain adaptation and knowledge generalization for vision and multimodal models.
Applied machine learning Remaining useful life (RUL) prediction, state of charge (SOC) prediction, AI for Science.

Services

2024 **Reviewer,** IEEE TNNLS, IEEE TMM, IEEE JBHI, AAAI, NeurIPS, ICLR
2025 **Reviewer,** IEEE TPAMI, IEEE TNNLS, IEEE TCSVT, NeurIPS, ICCV
2026 **Reviewer,** ICML, Neural Networks, IEEE TCSVT, IJCV